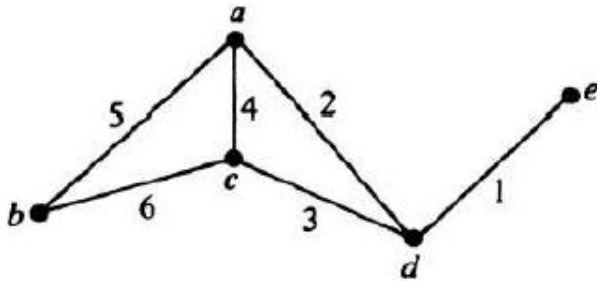


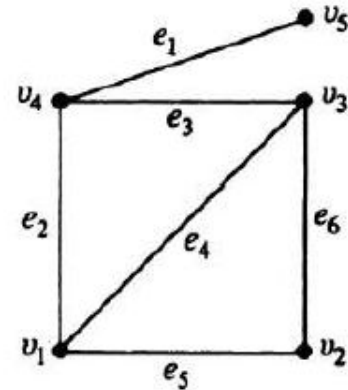
Paths and Circuits

ISOMORPHISM

- Two graphs are thought of as equivalent (and called isomorphic) if they have identical behavior in terms of graph-theoretic properties
 - How they are drawn doesn't matter
- For two isomorphic graphs, there must be one-to-one correspondence between their vertices and edges while preserving incidence relationship



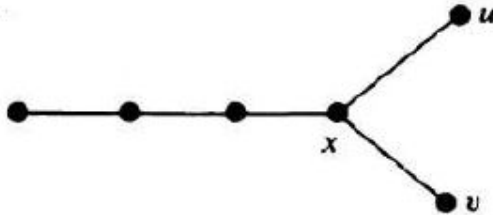
(a)



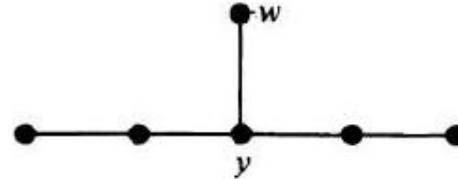
(b)

ISOMORPHISM

- It is immediately apparent by the definition of isomorphism that two isomorphic graphs must have
 - The same number of vertices.
 - The same number of edges.
 - An equal number of vertices with a given degree.
- However, these conditions are by no means sufficient, as we can see for the two graphs below



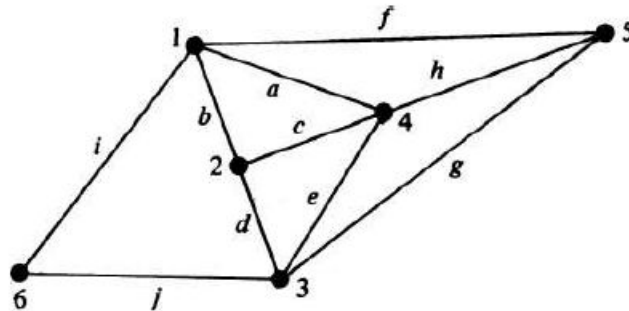
(a)



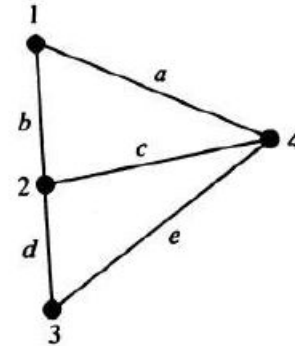
(b)

SUBGRAPHS

- The concept of subgraph is akin to the concept of subset in set theory. A subgraph can be thought of as being contained in (or a part of) another graph.
 - The symbol from set theory, $g \subset G$, is used in stating “ g is a subgraph of G .”



(a)



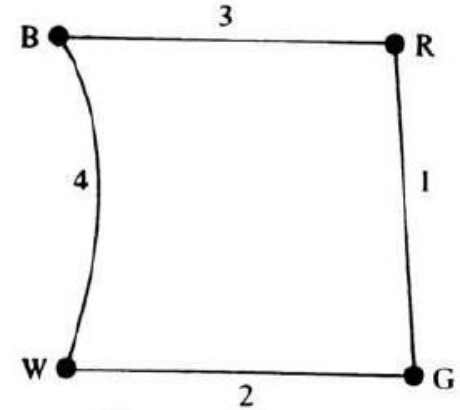
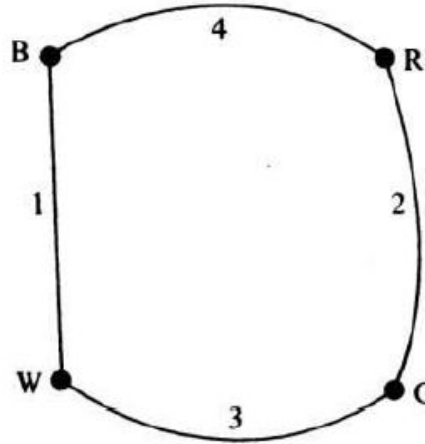
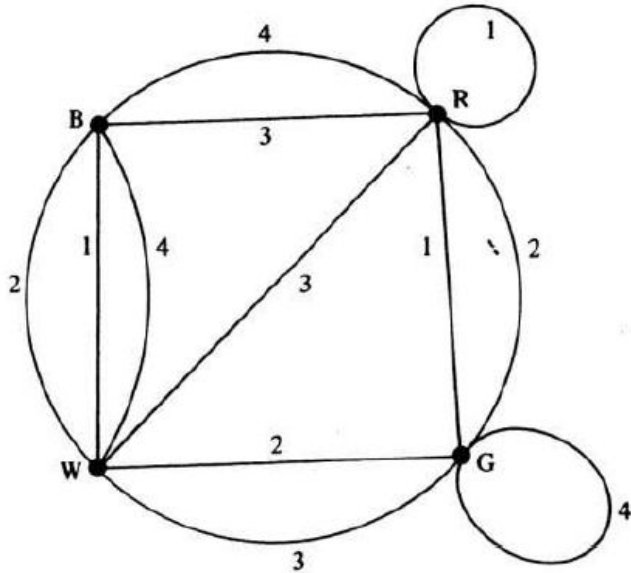
(b)

SUBGRAPHS

- The following observations can be made immediately:
 - Every graph is its own subgraph.
 - A subgraph of a subgraph of G is a subgraph of G .
 - A single vertex in a graph G is a subgraph of G .
 - A single edge in G , together with its end vertices, is also a subgraph of G

SUBGRAPHS

- Two (or more) subgraphs g_1 and g_2 of a graph G are said to be edge disjoint if g_1 and g_2 do not have any edges in common.



INSTANT INSANITY PUZZLE

This is an example of how a problem can be solved using graphs.

It will involve two steps:

- First, convert the problem into a problem of graph theory
- Second, solve the problem using graph theory

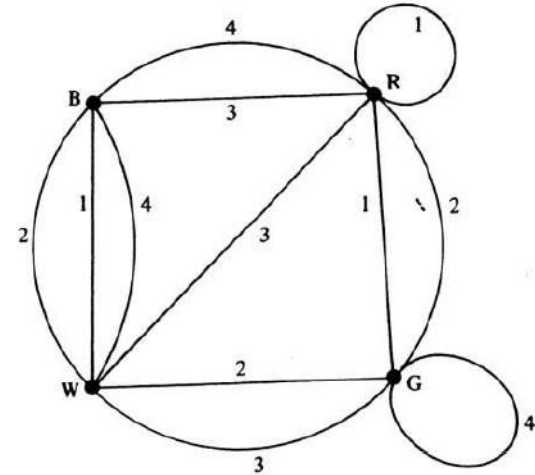
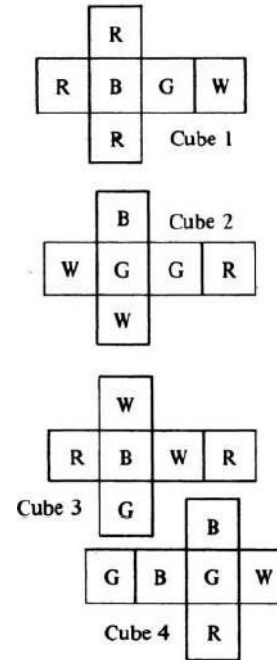
Problem: We are given four cubes. The six faces of every cube are variously colored blue, green, red, or white. Is it possible to stack the cubes one on top of another to form a column such that no color appears twice on any of the four sides of this column?

INSTANT INSANITY PUZZLE

Step 1: Draw a graph with four vertices B, G, R, and W, one for each color. Put label 1 on all three edges resulting from cube 1.

A self-loop with the edge labeled 1 at vertex R, for instance, would result if cube 1 had a pair of opposite faces both colored red.

Repeat the procedure for the other three cubes one by one on the same graph

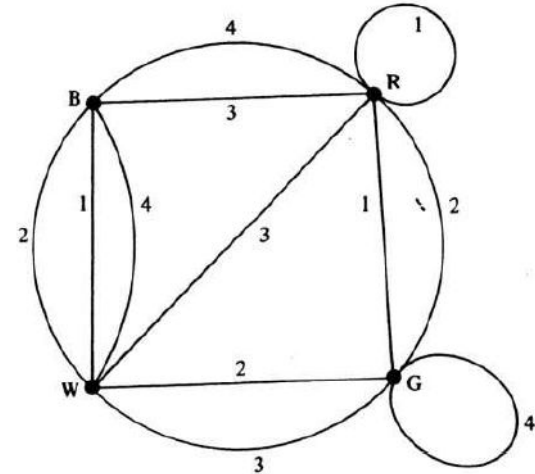
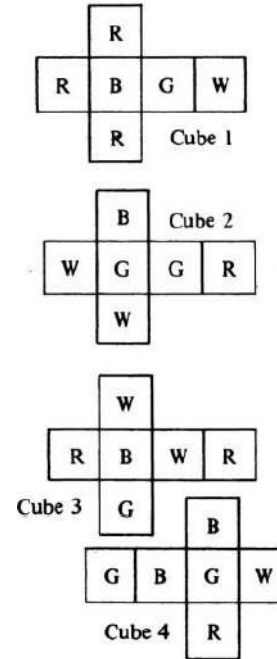


INSTANT INSANITY PUZZLE

Step 2: The degree of each vertex is the total number of faces with the corresponding color.

Consider two opposite vertical sides of the desired column of four cubes, say facing north and south. A subgraph (with four edges) will represent these eight faces—four facing south and four north. Each of the four edges in this subgraph will have a different label—1, 2, 3, and 4.

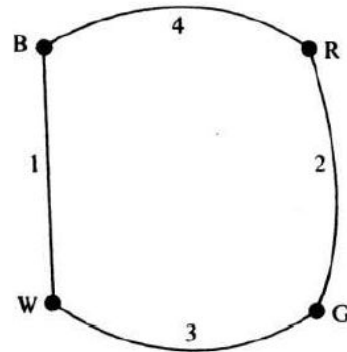
Moreover, no color occurs twice on either the north side or south side of the column if and only if every vertex in this subgraph is of degree two.



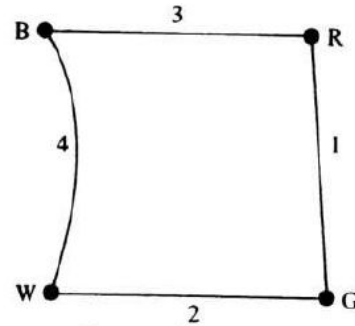
INSTANT INSANITY PUZZLE

Step 2: Exactly the same argument applies to the other two sides, east and west, of the column.

Thus the four cubes can be arranged (to form a column such that no color appears more than once on any side) if and only if there exist two edge-disjoint subgraphs, each with four edges, each of the edges labeled differently, and such that each vertex is of degree two.



(a) North-South Subgraph

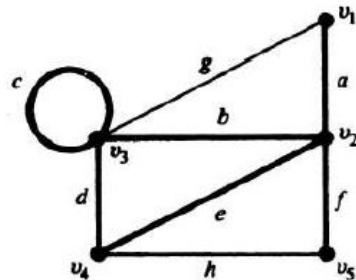


(b) East-West Subgraph

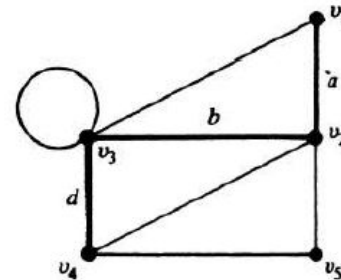
WALKS, PATHS, AND CIRCUITS

A walk is defined as a finite alternating sequence of vertices and edges, beginning and ending with vertices, such that each edge is incident with the vertices preceding and following it. No edge appears (is covered or traversed) more than once in a walk. A vertex, however, may appear more than once.

For instance, in the following graph, $v_1 a v_2 b v_3 c v_3 d v_4 e v_2 f v_5$ is a walk shown with heavy lines.



(a) An Open Walk



(b) A Path of Length Three

WALKS, PATHS, AND CIRCUITS

Vertices with which a walk begins and ends are called its terminal vertices.

It is possible for a walk to begin and end at the same vertex. Such a walk is called a **closed walk**. A walk that is not closed (i.e., the terminal vertices are distinct) is called an **open walk**. An open walk in which **no vertex appears more than once** is called a **path** (or a simple path or an elementary path).

A closed walk in which no vertex (except the initial and the final vertex) appears more than once is called a **circuit**.

CONNECTED GRAPHS, DISCONNECTED GRAPHS, AND COMPONENTS

A graph G is said to be **connected** if there is at least one path between every pair of vertices in G . Otherwise, G is **disconnected**. A null graph of more than one vertex is disconnected.

A disconnected graph consists of two or more connected graphs. Each of these connected subgraphs is called a **component**.

THEOREM 2-2

“If a graph (connected or disconnected) has exactly two vertices of odd degree, there must be a path joining these two vertices.”

Let G be a graph with all even vertices (vertices of even degree) except vertices v_1 and v_2 , which are odd.

From **Theorem 1-1**, which holds for every graph and therefore for every component of a disconnected graph, no graph can have an odd number of odd vertices. Therefore, in graph G , v_1 and v_2 must belong to the same component, and hence must have a path between them.

THEOREM 2-3

“A simple graph (i.e., a graph without parallel edges or self-loops) with n vertices and k components can have at most $(n - k)(n - k + 1)/2$ edges.”

Let the number of vertices in each of the k components of a graph G be $n_1, n_2, \dots, n_k$. Thus we have

$$n_1 + n_2 + \dots + n_k = n$$

$$n_i \geq 1$$

For the i th component, the maximum number of edges is

$$\frac{n_i (n_i - 1)}{2}$$

THEOREM 2-3

$$\sum_{i=1}^k (n_i - 1) = n - k$$

Squaring both sides,

$$\sum_{i=1}^k n_i^2 - 2 \sum_{i=1}^k n_i + k + C = n^2 - 2nk + k^2$$

$$\sum_{i=1}^k n_i^2 \leq n^2 - 2nk + k^2 + 2n - k$$

THEOREM 2-3

Since each component can have maximum edges as

$$\frac{n_i (n_i - 1)}{2}$$

So, maximum edges possible in whole graph G is

$$\begin{aligned} \sum_{i=1}^k \frac{n_i (n_i - 1)}{2} &= \frac{1}{2} \left(\sum_{i=1}^k n_i^2 - n \right) \\ &\leq \frac{1}{2} (n^2 - 2nk + k^2 + 2n - k - n) \\ &\leq \frac{1}{2} [(n - k)^2 + (n - k)] \\ &\leq \frac{1}{2} (n - k) (n - k + 1) \end{aligned}$$

EULER GRAPHS

In what type of graph G is it possible to find a closed walk running through every edge of G exactly once?

Such a walk is called an **Euler line**, and a graph that consists of an Euler line is called an **Euler graph**.

THEOREM 2-4

“A given connected graph G is an Euler graph if and only if all vertices of G are of even degree.”

Notice that the theorem statement contains the phrase “**if and only if**”. For proving statements like these, we need to prove both the necessity and sufficiency of it. That is, we need to prove it **bidirectionally**.

At first, we prove one direction, where we start with the premise that G is an Euler graph. Since it is an Euler Graph, it therefore contains an Euler line (which is a closed walk). In tracing this walk we observe that every time the walk meets a vertex v it goes through two “new” edges incident on v —with one we “entered” v and with the other “exited.”

THEOREM 2-4

“A given connected graph G is an Euler graph if and only if all vertices of G are of even degree.”

This is true not only of all intermediate vertices of the walk but also of the terminal vertex, because we “exited” and “entered” the same vertex at the beginning and end of the walk, respectively.

Thus if G is an Euler graph, the degree of every vertex is even.

THEOREM 2-4

“A given connected graph G is an Euler graph if and only if all vertices of G are of even degree.”

Now we prove it from the other direction, where we start with the premise that all vertices of G are of even degree, and we have to prove that G is an Euler Graph.

We construct a walk starting at an arbitrary vertex v and going through the edges of G such that no edge is traced more than once.

Since every vertex is of even degree, we can exit from every vertex we enter; the tracing cannot stop at any vertex but v . And since v is also of even degree, we shall eventually reach v when the tracing comes to an end.

THEOREM 2-4

“A given connected graph G is an Euler graph if and only if all vertices of G are of even degree.”

If this closed walk h we just traced includes all the edges of G , G is an Euler graph. If not, we remove from G all the edges in h and obtain a subgraph h' of G formed by the remaining edges.

Since both G and h have all their vertices of even degree, the degrees of the vertices of h' are also even. Moreover, h' must touch h at least at one vertex a , because G is connected.

THEOREM 2-4

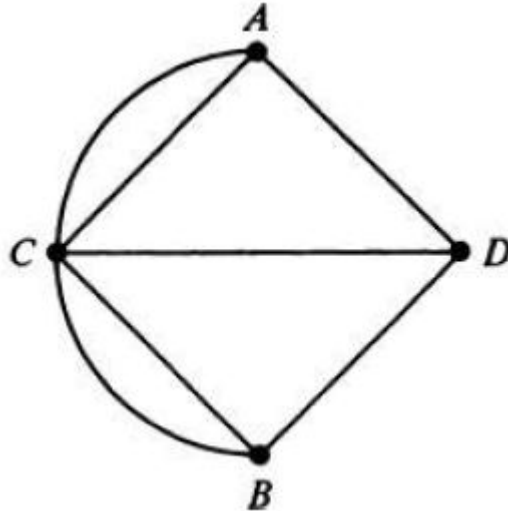
“A given connected graph G is an Euler graph if and only if all vertices of G are of even degree.”

Starting from a , we can again construct a new walk in graph h' . Since all the vertices of h' are of even degree, this walk in h' must terminate at vertex a ; but this walk in h' can be combined with h to form a new walk.

This process can be repeated until we obtain a closed walk that traverses all the edges of G . Thus G is an Euler graph.

Königsberg Bridge Problem

Looking at the graph of the Königsberg bridges, we find that not all its vertices are of even degree. Hence, it is not an Euler graph. Thus it is not possible to walk over each of the seven bridges exactly once and return to the starting point.

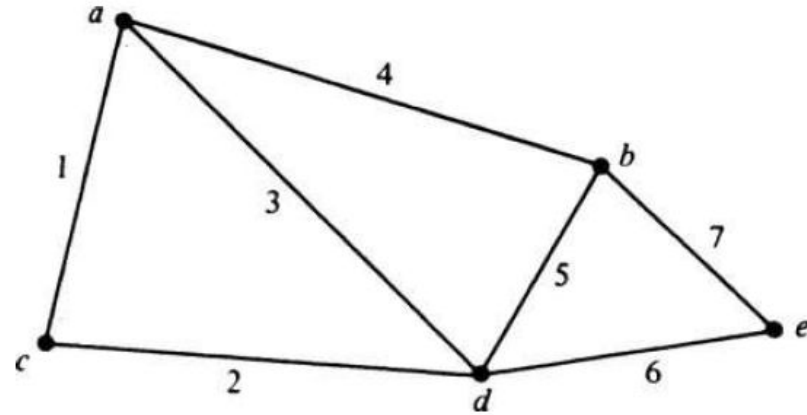


UNICURSAL GRAPH

In defining an Euler line, we can drop the requirement that the walk be closed.

For example, the walk $a1c2d3a4b5d6e7b$ includes all the edges of the graph and does not retrace any edge, but is not closed (does not come back to the starting vertex).

We shall call such an open walk a unicursal line or an open Euler line. A connected graph that has a unicursal line will be called a unicursal graph.



GRAPH OPERATIONS

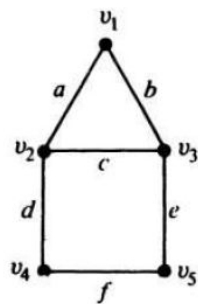
As is the case with most mathematical entities, it is convenient to consider a large graph as a combination of small ones and to derive its properties from those of the small ones.

The union of two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is another graph $G_3 = G_1 \cup G_2$ whose vertex set $V_3 = V_1 \cup V_2$ and the edge set $E_3 = E_1 \cup E_2$.

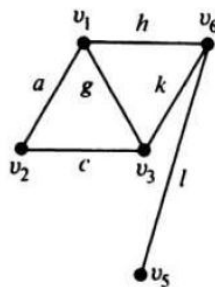
Likewise, the intersection $G_1 \cap G_2$ is a graph G_4 consisting only of those vertices and edges that are in both G_1 and G_2 .

The ring sum of two graphs G_1 and G_2 (written as $G_1 \oplus G_2$) is a graph consisting of the vertex set $V_1 \cup V_2$ and of edges that are either in G_1 or G_2 , but not in both (similar to XOR).

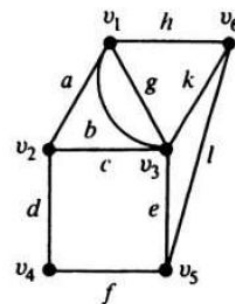
GRAPH OPERATIONS



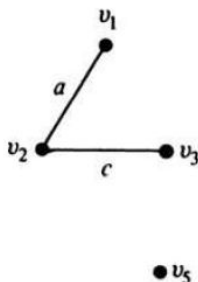
G_1



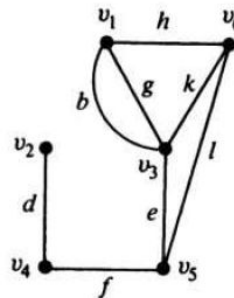
G_2



$G_1 \cup G_2$



$G_1 \cap G_2$



$G_1 \oplus G_2$

GRAPH OPERATIONS

Some Observations...

If G_1 and G_2 are edge disjoint, then $G_1 \cap G_2$ is a null graph, and $G_1 \oplus G_2 = G_1 \cup G_2$.
If G_1 and G_2 are vertex disjoint, then $G_1 \cap G_2$ is empty.

If g is a subgraph of G , then $G \oplus g$ is, by definition, that subgraph of G which remains after all the edges in g have been removed from G . Therefore, $G \oplus g$ is written as $G - g$, whenever $g \subseteq G$. Because of this complementary nature, $G \oplus g = G - g$ is often called the complement of g in G .

GRAPH OPERATIONS

Decomposition: A graph G is said to have been decomposed into two subgraphs g_1 and g_2 if $g_1 \cup g_2 = G$, and $g_1 \cap g_2 =$ a null graph.

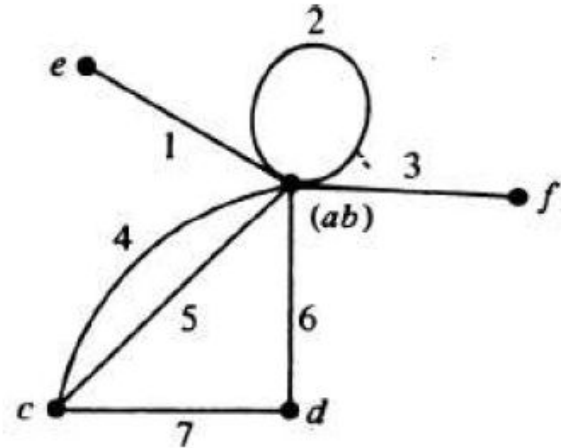
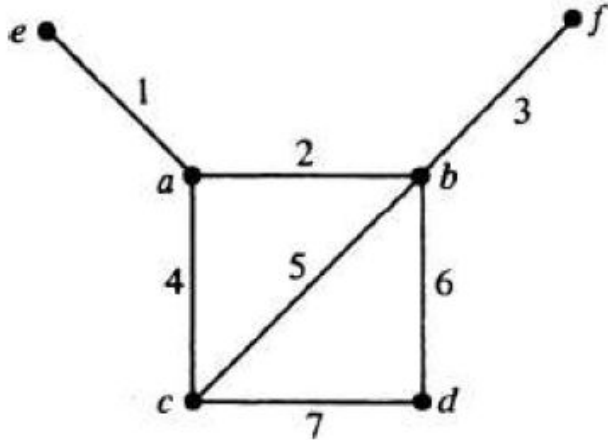
A graph containing m edges $\{e_1, e_2, \dots, e_m\}$ can be decomposed in $2^{m-1} - 1$ different ways into pairs of subgraphs g_1, g_2

Deletion: If v_i is a vertex in graph G , then $G - v_i$ denotes a subgraph of G obtained by deleting (i.e., removing) v_i from G . Deletion of a vertex always implies the deletion of all edges incident on that vertex.

On the otherhand, if e_j is an edge in G , then $G - e_j$ is a subgraph of G obtained by deleting e_j from G . Deletion of an edge does not imply deletion of its end vertices.

GRAPH OPERATIONS

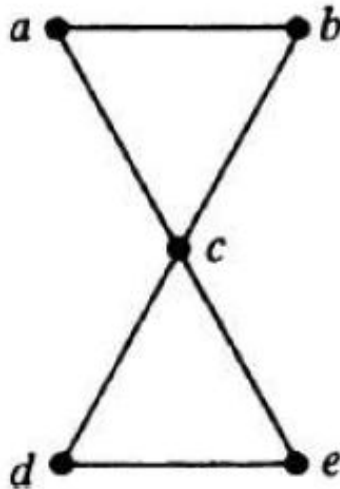
Fusion: A pair of vertices a, b in a graph are said to be fused (merged or identified) if the two vertices are replaced by a single new vertex such that every edge that was incident on either a or b or on both is incident on the new vertex. Thus fusion of two vertices does not alter the number of edges, but it reduces the number of vertices by one.



Arbitrarily Traceable Graphs

Suppose, for the following Euler Graph, we start from vertex a and trace the path $a b c$.

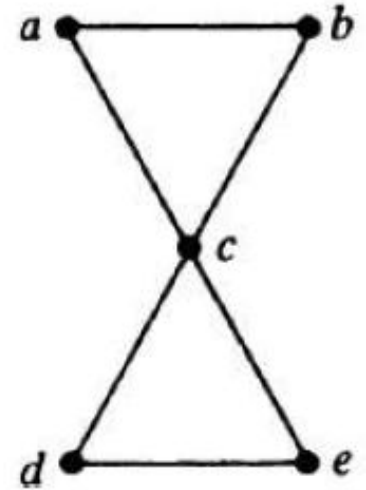
Now at c we have the choice of going to a , d , or e . If we took the first choice, we would only trace the circuit $a b c a$, which is not an Euler line. Thus, starting from a , we cannot trace the entire Euler line simply by moving along any edge that has not already been traversed.



Arbitrarily Traceable Graphs

What property must a vertex v in an Euler graph have such that an Euler line is always obtained when one follows any walk from vertex v according to the single rule that whenever one arrives at a vertex one shall select any edge (which has not been previously traversed)?

Such a graph is called an arbitrarily traceable graph from vertex v . For instance, this graph is an arbitrarily traceable graph from vertex c , but not from any other vertex.



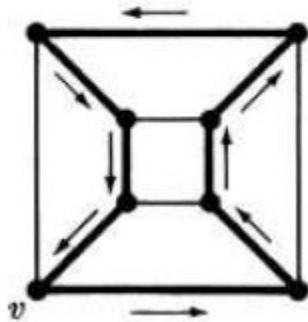
THEOREM 2-7

“An Euler graph G is arbitrarily traceable from vertex v in G if and only if every circuit in G contains v .”

For proof, interested students are given a link to a paper (found in the Narsingh Deo Book)

HAMILTONIAN PATHS AND CIRCUITS

A Hamiltonian circuit in a connected graph is defined as a closed walk that traverses **every vertex of G exactly once**, except of course the starting vertex, at which the walk also terminates (so only this is traversed twice).



HAMILTONIAN PATHS AND CIRCUITS

A Hamiltonian circuit in a graph of **n vertices** consists of exactly **n edges**.

What is a necessary and sufficient condition for a connected graph G to have a Hamiltonian circuit?

This problem, first posed by the famous Irish mathematician Sir William Rowan Hamilton in 1859, is still unsolved.

HAMILTONIAN PATHS AND CIRCUITS

The resemblance between the problem of an Euler line and that of a Hamiltonian circuit is deceptive. The latter is infinitely more complex.

Although one can find Hamiltonian circuits in many specific graphs, such as those shown earlier, there is no known criterion we can apply to determine the existence of a Hamiltonian circuit in general.

There are, however, certain types of graphs that always contain Hamiltonian circuits.

HAMILTONIAN PATHS AND CIRCUITS

If we remove any one edge from a Hamiltonian circuit, we are left with a path. This path is called a **Hamiltonian path**. Clearly, a Hamiltonian path in a graph G traverses every vertex of G , except it does not need to return to the initial vertex.

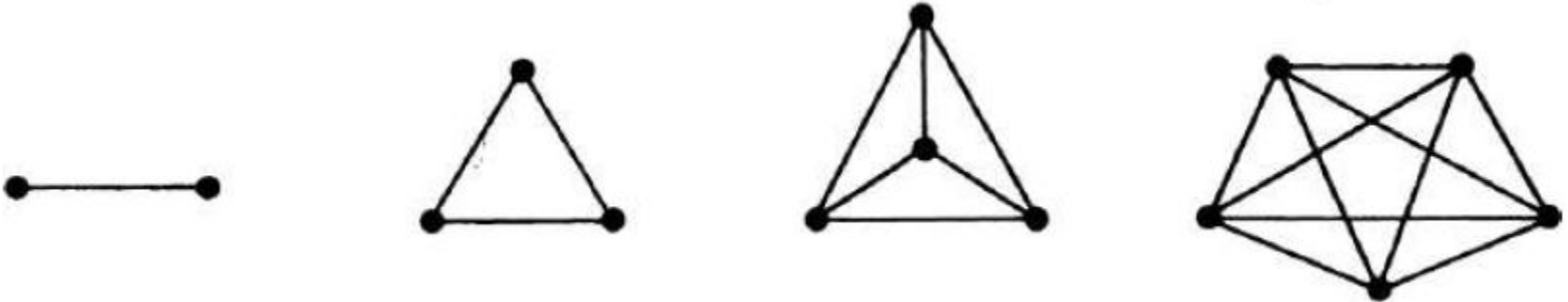
In considering the existence of a Hamiltonian circuit (or path), we need only consider simple graphs. This is because a Hamiltonian circuit (or path) traverses every vertex exactly once. Hence it cannot include a self-loop or a set of parallel edges.

Thus a general graph may be made simple by removing parallel edges and self-loops before looking for a Hamiltonian circuit in it.

HAMILTONIAN PATHS AND CIRCUITS

Now, what general class of graphs is guaranteed to have a Hamiltonian circuit? Complete graphs of three or more vertices constitute one such class, which we will focus on here.

Complete Graph: A simple graph in which there exists an edge between every pair of vertices is called a complete graph. Complete graphs of two, three, four, and five vertices are shown.



HAMILTONIAN PATHS AND CIRCUITS

It is easy to construct a Hamiltonian circuit in a complete graph of n vertices.

Let the vertices be numbered $v_1, v_2, \dots, v_n$. Since an edge exists between any two vertices, we can start from v_1 and traverse to v_2 , and v_3 , and so on to v_n , and finally from v_n to v_1 . This is a Hamiltonian circuit.

Number of Hamiltonian Circuits in a Graph

A given graph may contain more than one Hamiltonian circuit. Of interest are all the edge-disjoint Hamiltonian circuits in a graph.

The determination of the exact number of edge-disjoint Hamiltonian circuits (or paths) in a general graph is also an unsolved problem.

However, the number of edge-disjoint Hamiltonian circuits in a complete graph with **odd number of vertices** can be determined, as given by the following theorem.

THEOREM 2-8

“In a complete graph with n vertices there are $(n - 1)/2$ edge-disjoint Hamiltonian circuits, if n is an odd number ≥ 3 .”

Proof: A complete graph G of n vertices has $n(n - 1)/2$ edges, and each Hamiltonian circuit in G consists of n edges. Therefore, the number of edge-disjoint Hamiltonian circuits in G cannot exceed $(n - 1)/2$.

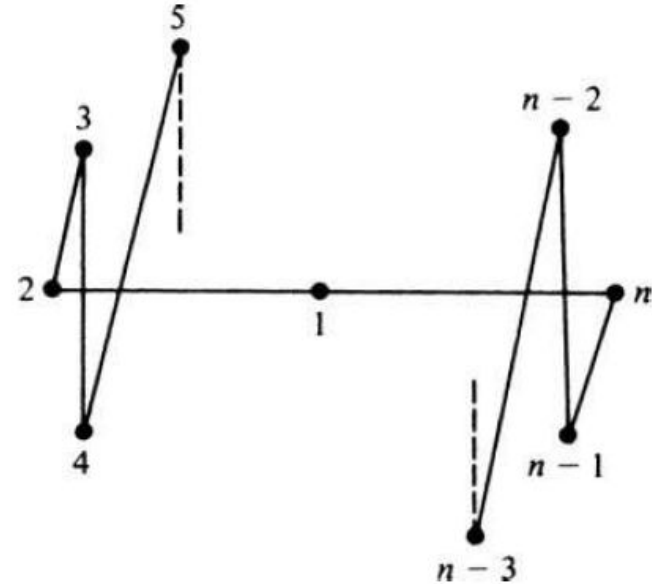
These $(n - 1)/2$ circuits can be constructed following this process discussed shortly.

THEOREM 2-8

Keeping the vertices fixed on a circle, rotate the polygonal pattern clockwise by $360/(n - 1)$, $2 \cdot 360/(n - 1)$, $3 \cdot 360/(n - 1)$, ..., $((n - 3)/2) \cdot 360/(n - 1)$ degrees.

Observe that each rotation produces a Hamiltonian circuit that has no edge in common with any of the previous ones. Thus we have $(n - 3)/2$ new Hamiltonian circuits, which along with the starting configuration gives us

$$\frac{n - 3}{2} + 1 = \frac{n - 1}{2}$$



SEATING ARRANGEMENT PROBLEM

Theorem 2-8 enables us to solve the problem of the seating arrangement at a round table, introduced in Chapter 1:

Since every member is allowed to sit next to any other member, G is a complete graph of nine vertices—nine being the number of people to be seated around the table. Every Seating arrangement around the table is clearly a Hamiltonian circuit.

The first day of their meeting they can sit in any order, and it will be a Hamiltonian circuit H_1 . The second day, if they are to sit such that every member must have different neighbors, we have to find another Hamiltonian circuit H_2 in G , with an entirely different set of edges from those in H_1 ; that is, H_1 and H_2 are edge-disjoint Hamiltonian circuits.

From Theorem 2-8 the number of edge-disjoint Hamiltonian circuits in G is four; therefore, only four such arrangements exist among nine people.

THEOREM 2-9

“A sufficient (but by no means necessary) condition for a simple graph G to have a Hamiltonian circuit is that the degree of every vertex in G be at least $n/2$, where n is the number of vertices in G .”

For proof, the interested students are referred to the original paper by Dirac (link given in the book)

TRAVELING-SALESMAN PROBLEM

A problem closely related to the question of Hamiltonian circuits is the traveling-salesman problem, stated as follows:

A salesman is required to visit a number of cities during a trip. Given the distances between the cities, in what order should he travel so as to visit every city precisely once and return home, with the minimum mileage traveled?

In this problem, if each of the cities has a road to every other city, we have a complete weighted graph. This graph has numerous Hamiltonian circuits, and we are to pick the one that has the smallest sum of distances

TRAVELING-SALESMAN PROBLEM

The total number of different (not edge disjoint, of course) Hamiltonian circuits in a complete graph of n vertices can be shown to be $(n - 1)!/2$.

This follows from the fact that starting from any vertex we have $n - 1$ edges to choose from the first vertex, $n - 2$ from the second, $n - 3$ from the third, and so on.

These being independent choices, we get $(n - 1)!$ possible number of choices.

This number is, however, divided by 2, because each Hamiltonian circuit has been counted twice.